Table 1: Bayes AUROC minus fixed-method AUROC for the six conceptual representatives in all 40 original settings with validated Bayes references. Lower regret is better.

One-sample Scenario	Bayes AUROC	Logistic correction	nnPU logistic	Naive CatBoost	CS CatBoost	RN CatBoost	PU gradient boosting
Reference	0.738	0.004	0.004	0.014	0.014	0.012	0.031
Prevalence $\pi = 0.01$	0.738	0.035	0.035	0.087	0.084	0.046	0.172
Prevalence $\pi = 0.05$	0.738	0.011	0.011	0.032	0.032	0.020	0.097
Prevalence $\pi = 0.10$	0.738	0.005	0.005	0.018	0.018	0.014	0.063
Prevalence $\pi = 0.15$	0.738	0.003	0.003	0.014	0.013	0.011	0.043
Prevalence $\pi = 0.25$	0.738	0.003	0.003	0.012	0.012	0.011	0.024
Prevalence $\pi = 0.30$	0.738	0.003	0.003	0.011	0.011	0.010	0.021
Prevalence $\pi = 0.35$	0.738	0.004	0.004	0.012	0.012	0.011	0.020
Prevalence $\pi = 0.40$	0.738	0.002	0.002	0.010	0.010	0.009	0.017
Prevalence $\pi = 0.45$	0.738	0.003	0.004	0.011	0.011	0.010	0.017
Prevalence $\pi = 0.50$	0.738	0.004	0.005	0.012	0.013	0.010	0.018
Labeling probability $c = 0.05$	0.738	0.020	0.021	0.052	0.050	0.027	0.122
Labeling probability $c = 0.10$	0.738	0.012	0.012	0.032	0.031	0.020	0.090
Labeling probability $c = 0.15$	0.738	0.008	0.008	0.022	0.022	0.016	0.066
Labeling probability $c = 0.20$	0.738	0.005	0.005	0.017	0.017	0.014	0.052
Labeling probability $c = 0.25$	0.738	0.005	0.005	0.016	0.015	0.012	0.038
Labeling probability $c = 0.35$	0.738	0.003	0.003	0.011	0.011	0.010	0.023
Labeling probability $c = 0.40$	0.738	0.003	0.003	0.011	0.011	0.011	0.020
Labeling probability $c = 0.45$	0.738	0.003	0.003	0.011	0.011	0.011	0.019
Labeling probability $c = 0.50$	0.738	0.003	0.003	0.010	0.009	0.010	0.016
Class separation $\delta = 0.50$	0.638	0.007	0.007	0.017	0.018	0.013	0.034
Class separation $\delta = 0.70$	0.690	0.003	0.003	0.014	0.014	0.011	0.031
Class separation $\delta = 1.20$	0.802	0.003	0.003	0.012	0.011	0.010	0.025
High dimension ($p = 50$)	0.738	0.016	0.016	0.016	0.017	0.011	0.036
High dimension ($p = 100$)	0.738	0.033	0.034	0.021	0.021	0.015	0.039
High dimension ($p = 500$)	0.738	0.095	0.105	0.025	0.025	0.019	0.043
Correlation $\rho = 0.30$	0.738	0.003	0.003	0.012	0.012	0.014	0.026
Correlation $\rho = 0.60$	0.738	0.003	0.003	0.018	0.017	0.015	0.036
Prevalence $\pi = 0.01$, $c = 0.50$	0.738	0.023	0.023	0.065	0.067	0.031	0.146
Prevalence $\pi = 0.05$, $c = 0.10$	0.738	0.025	0.025	0.070	0.070	0.034	0.150
Prevalence $\pi = 0.05$, $c = 0.50$	0.738	0.006	0.006	0.020	0.019	0.014	0.068
Prevalence $\pi = 0.10$, $c = 0.05$	0.738	0.031	0.032	0.080	0.085	0.039	0.151
Prevalence $\pi = 0.10$, $c = 0.10$	0.738	0.015	0.015	0.043	0.040	0.022	0.113
Prevalence $\pi = 0.10$, $c = 0.50$	0.738	0.003	0.003	0.013	0.012	0.011	0.033
Prevalence $\pi = 0.50$, $c = 0.05$	0.738	0.022	0.030	0.051	0.052	0.031	0.088
Prevalence $\pi = 0.50$, $c = 0.10$	0.738	0.010	0.012	0.028	0.028	0.019	0.055
Prevalence $\pi = 0.50$, $c = 0.50$	0.738	0.002	0.002	0.008	0.008	0.008	0.011
Curved boundary	0.738	0.036	0.036	0.032	0.031	0.043	0.050
Feature interaction	0.739	0.080	0.080	0.026	0.025	0.126	0.087
Unequal class covariance	0.739	0.027	0.027	0.008	0.009	0.052	0.026

Two-sample Scenario	Bayes AUROC	Logistic correction	nnPU logistic	Naive CatBoost	CS CatBoost	RN CatBoost	PU gradient boosting
Reference	0.738	0.004	0.004	0.014	0.013	0.012	0.023
Prevalence $\pi = 0.01$	0.738	0.004	0.004	0.011	0.011	0.011	0.078
Prevalence $\pi = 0.05$	0.738	0.002	0.002	0.010	0.010	0.010	0.038
Prevalence $\pi = 0.10$	0.738	0.003	0.003	0.012	0.011	0.010	0.028
Prevalence $\pi = 0.15$	0.738	0.003	0.003	0.013	0.013	0.011	0.025
Prevalence $\pi = 0.25$	0.738	0.005	0.005	0.015	0.016	0.013	0.024
Prevalence $\pi = 0.30$	0.738	0.006	0.006	0.016	0.017	0.014	0.025
Prevalence $\pi = 0.35$	0.738	0.006	0.007	0.018	0.018	0.015	0.028
Prevalence $\pi = 0.40$	0.738	0.009	0.009	0.021	0.022	0.017	0.031
Prevalence $\pi = 0.45$	0.738	0.009	0.011	0.023	0.024	0.017	0.033
Prevalence $\pi = 0.50$	0.738	0.012	0.015	0.028	0.029	0.021	0.038
Labeled positives $n_P = 50$	0.738	0.019	0.020	0.054	0.052	0.026	0.064
Labeled positives $n_P = 100$	0.738	0.011	0.011	0.032	0.032	0.021	0.045
Labeled positives $n_P = 150$	0.738	0.008	0.008	0.023	0.023	0.017	0.034
Labeled positives $n_P = 200$	0.738	0.006	0.006	0.019	0.018	0.014	0.029
Labeled positives $n_P = 250$	0.738	0.006	0.006	0.017	0.017	0.014	0.026
Labeled positives $n_P = 350$	0.738	0.004	0.004	0.013	0.013	0.011	0.023
Labeled positives $n_P = 400$	0.738	0.001	0.002	0.010	0.010	0.009	0.020
Labeled positives $n_P = 450$	0.738	0.003	0.003	0.011	0.011	0.010	0.020
Labeled positives $n_P = 500$	0.738	0.002	0.003	0.010	0.010	0.009	0.020
Class separation $\delta = 0.50$	0.638	0.005	0.005	0.018	0.019	0.014	0.025
Class separation $\delta = 0.70$	0.690	0.004	0.005	0.015	0.015	0.012	0.023
Class separation $\delta = 1.20$	0.802	0.004	0.004	0.013	0.012	0.011	0.024
High dimension ($p = 50$)	0.738	0.019	0.020	0.019	0.020	0.013	0.025
High dimension ($p = 100$)	0.738	0.036	0.038	0.023	0.023	0.016	0.028
High dimension ($p = 500$)	0.738	0.104	0.107	0.030	0.030	0.022	0.035
Correlation $\rho = 0.30$	0.738	0.005	0.005	0.013	0.012	0.014	0.021
Correlation $\rho = 0.60$	0.738	0.004	0.004	0.020	0.019	0.016	0.030
Prevalence $\pi = 0.01$, $n_P = 25$	0.738	0.019	0.020	0.063	0.062	0.029	0.145
Prevalence $\pi = 0.01$, $n_P = 100$	0.738	0.004	0.004	0.022	0.021	0.013	0.103
Prevalence $\pi = 0.01$, $n_P = 500$	0.738	0.000	0.001	0.005	0.005	0.007	0.066
Prevalence $\pi = 0.10$, $n_P = 25$	0.738	0.032	0.033	0.078	0.083	0.040	0.091
Prevalence $\pi = 0.10$, $n_P = 100$	0.738	0.008	0.008	0.027	0.026	0.017	0.043
Prevalence $\pi = 0.10$, $n_P = 500$	0.738	0.003	0.003	0.009	0.009	0.010	0.026
Prevalence $\pi = 0.50$, $n_P = 25$	0.738	0.078	0.110	0.139	0.141	0.086	0.166
Prevalence $\pi = 0.50$, $n_P = 100$	0.738	0.028	0.038	0.060	0.070	0.037	0.068
Prevalence $\pi = 0.50$, $n_P = 500$	0.738	0.007	0.008	0.018	0.019	0.014	0.030
Curved boundary	0.738	0.035	0.035	0.033	0.032	0.042	0.043
Feature interaction	0.739	0.080	0.080	0.029	0.027	0.125	0.085
Unequal class covariance	0.739	0.028	0.028	0.010	0.010	0.052	0.019

Note. Method cells show Bayes AUROC minus observed AUROC; lower regret is better. Bayes references are the existing original benchmark analytic or exact-likelihood-ratio Monte Carlo values; Monte Carlo reference uncertainty is retained in original benchmark. All means use 100 repetitions. No new Bayes approximation is introduced.